types of semi-automatic sample preparation systems are commercially available. They are particularly useful in bio-medical tracer research, where often large numbers of samples are produced.

9.12. STATISTICS OF COUNTING AND ASSOCIATED ERROR

Even if the experimental design and execution are perfect so that the determinant error is eliminated in experiments involving radioactivity there is always a random error due to the statistical nature of radioactive decay. Each radioactive atom has a certain probability of decay within any one time interval. Consequently, since this probability allows unlikely processes to occur occasionally and likely processes not to occur in any particular time interval, the number of decays may be more or less than the number in another similar interval for the same sample. It is necessary, when counting a sample, to be able to calculate the probability that the recorded count rate is within certain limits of the true (or average) count rate.

The binomial distribution law correctly expresses this probability, but it is common practice to use either the Poisson distribution or the normal Gaussian distribution functions since both approximate the first but are much simpler to use. If the average number of counts is high (above 100) the Gaussian function may be used with no appreciable error. The probability for observing a measured value of total count N is

$$P(N) = (2\pi\overline{N})^{-½}\, e^{-½\left(\overline{N}-N\right)^2/\overline{N}} \tag{9.33}$$

The standard deviation σ is given in such cases by

$$\sigma = \sqrt{N} \tag{9.34}$$

In these equations $P(N)$ is the probability of the occurrence of the value N while $\overline{N}$ is the arithmetic mean of all the measured values and N is the measured value. Figure 9.25 shows the Poisson and Gaussian distributions for $\overline{N}= 20$ counts. The standard deviation ("statistical error"), according to the theory of errors, indicates a 68% probability that the measured value is within $\pm\sigma$ of the average "true" value $\overline{N}$. For 100 measured counts the value 100 ± 10 indicates that there is a 68% probability that the "true" value will be in the interval between 90 and 110 counts. If the error limit is listed as 2σ the probability that the "true" count will be between these limits is 95.5%; for 3σ it is 99.7%. Figure 9.26 shows the relationship between K, the number of standard deviations, and the probability that the true figure lies outside the limits expressed by K. For example, at $K = 1$ (i.e. the random error is $\pm1\sigma$) the Figure indicates that the probability of the true value being outside $N \pm \sigma$ is 0.32 (or 32%). This agrees with the observation that the probability is 68%, i.e. the "true" value is within the limits of $\pm1\sigma$. Figure 9.25 can be used to establish a "rule of thumb" for rejection of unlikely data. If any measurement

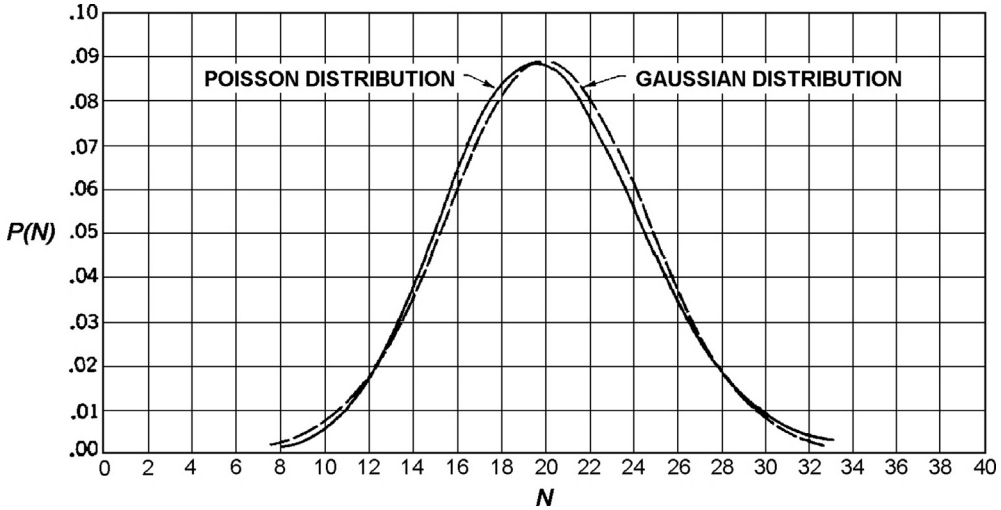


Figure 9.25 Poisson (smooth) and Gaussian (dashed line) distributions for $\overline{N} = 20$.

differs from the average value by more than five times the probable error it may be rejected as the probability is less than one in a thousand that this is a true random error. The probable error is the 50% probability which corresponds to 0.67σ.

From the relationship of σ and N it follows that the greater the number of collected counts the smaller the uncertainty. For high accuracy it is obviously necessary to obtain a large number of counts either by using samples of high radioactivity or by using long counting times.

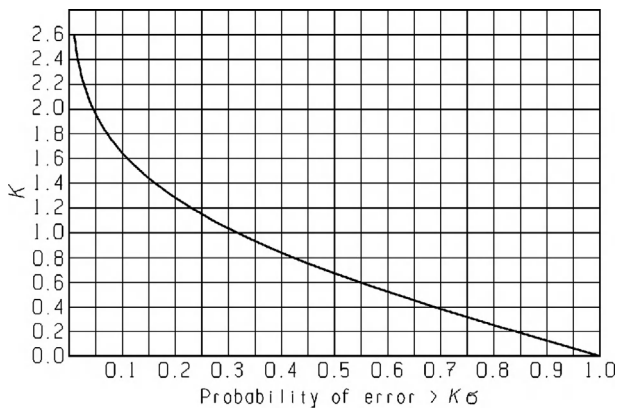


Figure 9.26 The probability that an error will be greater than $K\sigma$ for different K-values.

In order to obtain the value of the radioactivity of the sample, corrections must be made for background activity. If our measurements give $N \pm \sigma$ counts for the sample and $N_0 \pm \sigma_0$ for the background count, the correct value becomes

$$N_{\text{corr}} = (N - N_0) \pm \left(\sigma^2 + \sigma_0^2\right)^{1/2} \tag{9.35}$$

If the sample was counted for a time Δt and the background for a time Δt_0, the measured rate of radioactive decay is

$$R = N/\Delta t - N_0/\Delta t_0 \pm \left[(\sigma/\Delta t)^2 + (\sigma_0/\Delta t_0)^2\right]^{1/2} \tag{9.36}$$

It is extremely important in dealing with radioactivity to keep in mind at all times the statistical nature of the count rate. Every measured count has an uncertainty and the agreement between two counts can only be assessed in terms of the probability reflected in terms of σ.

The statistical nature of radioactive decay also leads to an uneven distribution of decays in time which is important when handling dead-time corrections and discussing required system time resolution. Let us first assume that a decay has occurred at time $t = 0$. What is then the differential probability that the next decay will take place within a short time interval, dt, after a time interval t has passed? Two independent processes must then occur in series. No decay may take place within the time interval from 0 to t, probability $P(0)$, followed by a decay within the time interval dt, probability rdt. The total probability is then given by the product of the individual probabilities of the two processes, i.e. if the combined probability is denoted $P(t)\mathrm{d}t$ then

$$P(t)\mathrm{d}t = P(0) \times r\ \mathrm{d}t \tag{9.37}$$

By applying the Poisson distribution and noting that r is equal to λN_0 we obtain

$$P(t)\mathrm{d}t = \lambda N_0\ \mathrm{e}^{-\lambda N_0 t}\ \mathrm{d}t = A_0\ \mathrm{e}^{-A_0 t}\ \mathrm{d}t \tag{9.38}$$

Equation (9.38) implies that the decays are not evenly distributed i time but often occur in clusters. The time interval between pairs of decays has indeed its maximum probability at zero.

In experimental work with radionuclides many other errors occur in addition to the statistical uncertainty in the count rate. Such errors may originate in the weighing or volumetric measurements, pH determination, etc. Such errors must also be considered in presenting the final results. For such composite errors, the *law of error propagation* must be applied:

$$\sigma_{\text{F}} = \left[(\sigma_{\text{A}}\ \mathrm{d}F/\mathrm{d}A)^2 + (\sigma_{\text{B}}\ \mathrm{d}F/\mathrm{d}B)^2\right]^{1/2} \tag{9.39}$$

where σ_{F} is one standard deviation in F, which is a function of the uncorrected variables A, B, …, with the standard errors σ_{A}, σ_{B}, etc. Commonly the unbiased estimation of the

standard deviation of σ is expressed as s. For n events, x_i, having the average $\overline{x}$, s is estimated according to:

$$s = \left\{ [1/(n-1)] \times \sum_{i=1}^{n} (x_i - \overline{x})^2 \right\}^{1/2} \tag{9.40}$$

For the product $A \times B$ and ratio A/B, one obtains

$$F = (A \times B)(1 \pm s) \tag{9.41}$$

and

$$F = (A/B)(1 \pm s) \tag{9.42}$$

where

$$s = \left[(\sigma_A/A)^2 + (\sigma_B/B)^2 \right]^{1/2} \tag{9.43}$$

For the function A^x and $\log A$ the following relations are valid

$$F = A^x \pm x \times A^{x-1}\, \sigma_A \tag{9.44}$$

$$F = \log A \pm [\sigma_A/(2.303A)]^{1/2} \tag{9.45}$$

A useful technique for checking that the error in the measurements has a Gaussian distribution is the so-called "χ-square" test. The quantity χ^2 is calculated from

$$\chi^2 = \left[\sum_{i=1}^{M} (\overline{F} - F_i)^2 \right] \Big/ [\overline{F}(k-1)] \tag{9.46}$$

where M is the number of measurements (e.g. points on a curve) for which the function F is (believed to be) valid. χ^2 would have a value 0.5—1.0 when the Gaussian fit exceeds 50%. k is the degree of freedom, i.e. M minus the number of determined variables. This relation is generally valid; for simple counting systems F is replaced by N, the number of counts in a given time interval, and $k = M$.

9.12.1. One-atom-at-a-time measurements

The difficulty to obtain good measuring statistics makes measurement of *e.g.* distribution coefficients (D) difficult with a small amount of atoms. Figure 9.27 shows the uncertainty of distribution coefficients based on 20 and 70 decay events, respectively, which may correspond to all atoms present given a short half-life, a sufficiently large Δt, and a detection efficiency of ~100%. Not surprisingly the uncertainty has a minimum at D = 1. Thus, if performing experiments where the heaviest elements are compared to lighter homologues it is favorable if the heavy elements are measured under conditions where D

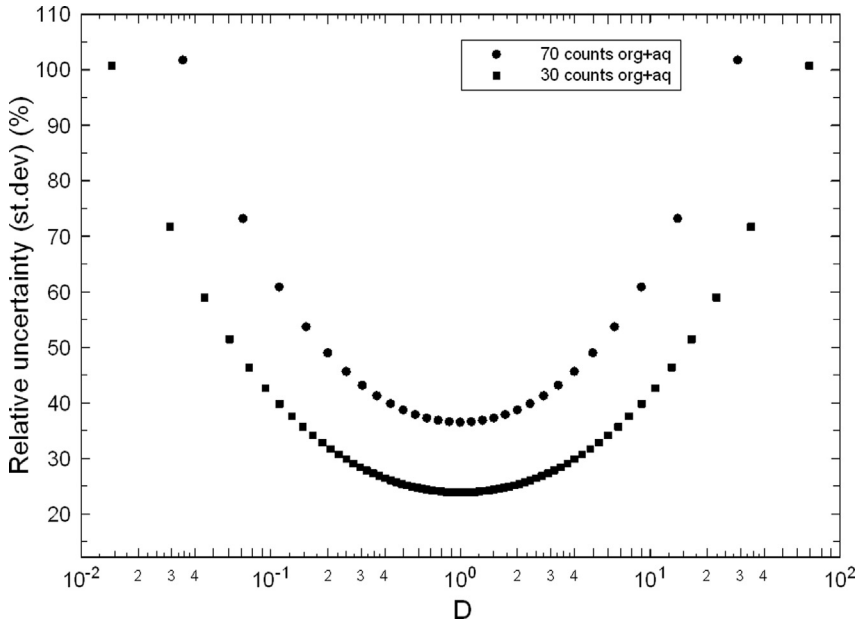


Figure 9.27 The relative size of one standard deviation in the D-value as a function of the D-value and a given total number of counts.

is not too far from 1 and the homologues, which can be produced in larger amounts, at a higher or lower D.

However, if fewer atoms than ~20 are detected other approaches are needed. One method is to calculate the probability for an atom to be present in the organic phase at a certain D-value. Then a random number is picked from some distribution, preferable a uniform distribution, thus assigning this particular atom to either the organic or aqueous phase. This may be repeated for the number of atoms desired by the aid of a simple computer program. If a large number of such calculations are performed they will approach the value of the uncertainty given above. Alternatively one may use the students T statistics to modify the multiple of the standard deviation needed for a given confidence interval.

When the total number of atoms used in an experiment is extremely small the usual fundamental thermodynamic descriptions no longer hold. Thus, the phase concept disappears when the medium contains only one or a few radioactive atoms. For reasons that are easily understood the law of mass action is also not applicable. If you, for example, only have one atom it is impossible for that atom to be in equilibrium with itself.

More specifically, in solvent extraction the distribution ratio between two phases looses its significance: a single atom can only exist in one of the two phases at the

moment of separation. An average distribution can, however, be obtained by performing a large number of identical experiments. The number of decay events in each phase is recorded. The decay statistics are replaced by the statistical distribution of experimental results.

Studies using extremely small amounts of any shortlived radionuclide involve a number of statistical considerations. The binomial law has to be used since the conditions for the Poisson distribution, *i.e.* that $N > 100$, and that the number of observed decays, $\overline{N}_{\Delta t}$, in the measuring time, Δt, is much smaller than N, are seldom fulfilled in this case. The probability, $P(1)$, of exactly one event happening is according to the binomial distribution:

$$P(1) = \left(\frac{n!}{(n-1)!}\right) p(1-p)^{(n-1)} \tag{9.47}$$

where p is the probability of decay, and n is the total number of atoms.

For a radioactive decay, the probability for exactly one decay during the time Δt in a sample containing 1 atom is thus:

$$p(\Delta t) = 1 - \mathrm{e}^{-\lambda \Delta t} \tag{9.49}$$

However, when the half-life is long compared to Δt it is possible to simplify this expression by a truncated Taylor series expansion to:

$$p(\Delta t) = \lambda \Delta t \tag{9.50}$$

In the general case involving N atoms, the following expressions for the expected average of decayed atoms, $\overline{N}_{\Delta t}$, and its variance, σ, are obtained:

$$\overline{N}_{\Delta t} = N\left(1 - \mathrm{e}^{-\lambda \Delta t}\right) \tag{9.51}$$

and

$$\sigma = \sqrt{\overline{N}_{\Delta t}}(1 - \mathrm{e}^{-\lambda \Delta t}) \tag{9.52}$$

9.13. EXERCISES

9.1. A detector has a 1 cm^2 efficient area perpendicular to a γ-particle flux produced by a source 7 m away. The sensitivity for the 0.73 MeV γ-radiation is 8.2%. (a) What must the source strength be for the detector to register 1000 cpm? (b) What fraction of the radiation is absorbed in the air space between source and detector?

9.2. A ~100 MeV fission fragment is stopped in a plastic plate with density ~1 and an average atomic spacing of 0.25 nm. Estimate (a) the range in the plate, and (b) the